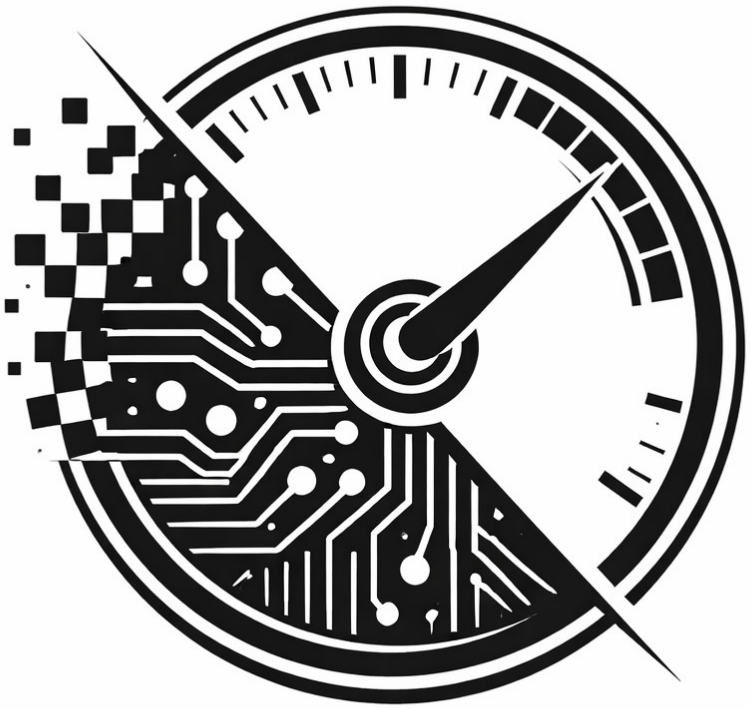
User manual

SWAP-CAN VOODOO T5

swap-parts.es



V0.9

Contents

Introduction....4
Technical data....4
Safety instructions....4
Intended use and compatibility....4
Electrical installation....5
Pinout....5
Power supply....6
CAN bus....6
Pressure sensors (oil and fuel)....7
Oil temperature sensor....7
AEM wideband sensor....8
Side/parking lights (+58)....8
SSR outputs....9
Mounting in the vehicle....10
Installing the base....10
Fitting and removing the gauge....10
First start-up....11
Symbol table....12
Using the gauge....12
Navigating between screens....12
Settings....12
Screens and data....13
Main....14
Data 2....15
Boost....16
Runtime....17
Sensors....18
AFR....19
Miscel....21
Alarms and SSR....22
Colour warning....22
Sustained knock alarm (>3s)....23
SSR output conditions....23

Logs..... 23
Recording a log..... 23
Log files..... 24
Analysing the logs..... 24
Diagnostic tables on the SD card..... 24
AFR Feedback..... 25
Knock Map..... 25
Knock_Count_ECU..... 25
Operation and clearing..... 25
Opening the tables..... 25
How it works..... 26
Update rate..... 26
Display and brightness..... 26
Updating..... 26
USB-C ports..... 26
Voodoo Updater..... 27
Care and storage..... 32
Messages and troubleshooting..... 32
Loading screen messages..... 32
Settings messages (SD and tables)..... 32
Audible signals..... 33
Common problems..... 33
Symbols and terms..... 33

Introduction

Congratulations on purchasing your Swap-CAN Voodoo. You now own a device **made by car enthusiasts, for car enthusiasts**.

These instructions are part of the product and contain important information about its safety and use, so please read them carefully and make sure you understand everything in them. Use the product only as described and within the stated limits. Pass on all documents if you hand the product over to someone else.

The Swap-CAN Voodoo is a data display for vehicles with **Saab Trionic 5** engine management. It connects to the ECU's CAN bus and shows live engine data on a round touchscreen, together with the readings from its own sensors (oil pressure, oil temperature and fuel pressure). It can also record logs, sound alarms, build diagnostic tables and switch up to four outputs (SSR) according to the conditions you choose.

Technical data

Feature	Value
Compatible engine management	Saab Trionic 5 (CAN bus at 615 kbit/s)
Supply through the 24-pin connector	5–17 V (recommended: vehicle 12 V)
Supply through USB-C	5 V (only when connected to a PC)
Maximum current	2 A (maximum rating of the wiring)
SSR outputs	4 outputs, 400 mA maximum each
Length of the 24-pin and 10-pin cables	approx. 15 cm (28 AWG wires)
Display	480 × 480 pixels, touchscreen, backlit
Operating temperature	0 °C to 50 °C
microSD card	up to 32 GB
Maximum file size	4 GB
Weight	92–94 g (gauge without the mounting base)

Safety instructions

- Do not reverse the positive and negative connections of the device.
- Do not get it wet, do not splash it and do not use it in high relative humidity.
- Do not connect or disconnect the 24-pin or 10-pin rear cables or connectors while the device is powered.
- Connect **all** the positive supply lines (marked **+15**) and **all** the ground lines (marked **GND**).
- Never supply it with more than **17 V**: this can damage the device.
- Do not connect loads of more than **400 mA** directly to an SSR output; use a relay.
- Do not open the data display unless strictly necessary, and do not knock the screen.
- Read this whole document before installation.

Intended use and compatibility

The Swap-CAN Voodoo works with **Saab Trionic 5** engine management units, because its CAN bus communication is specific to this ECU. These units were fitted to, among others, the following vehicles:

- Saab 9000 1994–1998 (turbo engines and some naturally aspirated ones)
- Saab 900 1994–1998 (turbo engines and some naturally aspirated ones)
- Saab 9-3 1998–2000

It is also suitable for any vehicle that has been fitted with an engine running a Trionic 5 ECU (engine swaps).

You need access to the **CAN H** and **CAN L** lines, which can be found at several points of the engine wiring of the vehicles listed above. Data acquisition runs on its own and needs no input from the user.

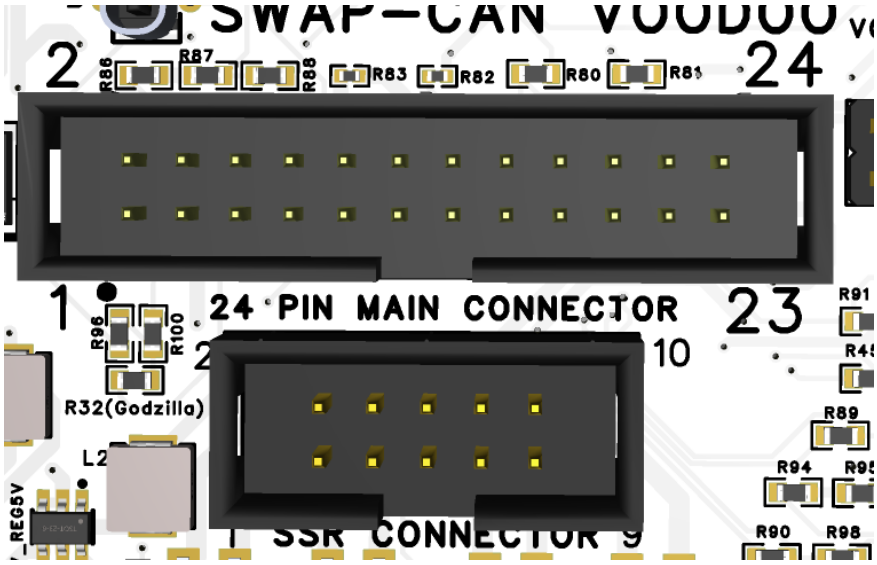
Important: some ECU software versions may not provide all the data. Some tuners also "remove" the symbol table, which makes it impossible for the Swap-CAN Voodoo to read it. In that case the table can be restored: please contact us.

Communication with the engine ECU is essential; without it no data will be shown. However, the gauge **can** run without the oil temperature, oil pressure or fuel pressure sensors: those values will simply not appear on screen (a "-" is shown instead) or in the logs.

Electrical installation

Pinout

The gauge has two connectors on the back: the **24-pin main connector** and the **10-pin SSR connector**. In the cables, **wire no. 1 is marked in red**; the next one is no. 2, the next no. 3, and so on. You have to count the wires starting from the red one.



Connectors seen from the back of the gauge. Odd-numbered pins are on the bottom row (notch side).

A printable pinout sheet with this same information is included with the product.

24-pin connector

Pin	Function	Pin	Function
1	+15 (supply)	13	Fuel pressure signal
2	+15 (supply)	14	Sensor GND
3	3.3 V output	15	CAN H
4	+15 (supply)	16	CAN H

Pin	Function	Pin	Function
5	EGT (in development, see note)	17	CAN L
6	IO9 (reserved)	18	CAN L
7	+58 side/parking lights (brightness dimming)	19	5 V sensor supply
8	IO3 (reserved)	20	5 V sensor supply
9	Oil temperature signal	21	5 V sensor supply
10	Sensor GND	22	GND
11	Oil pressure signal	23	GND
12	Sensor GND	24	GND

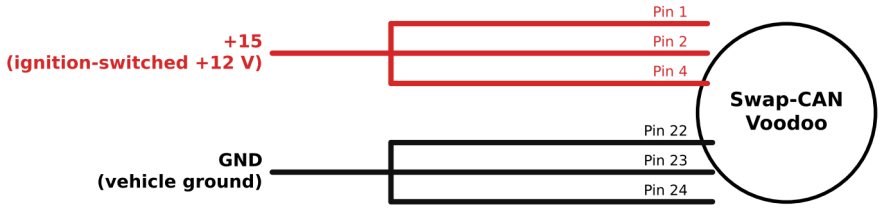
- **Pin 5 (EGT):** no function yet. It will require an external board that is currently in development. Do not connect.
- **Pins 6 (IO9) and 8 (IO3):** reserved for future functions. Do not connect.
- **Pin 7 (+58):** input from the side/parking lights circuit (DIN 72552 terminal 58, live whenever the side lights are switched on). See "Side/parking lights (+58)" below.
- **Pins 19, 20 and 21 (5 V sensor supply):** power supply for the pressure sensors.

10-pin SSR connector

Pin	Function	Pin	Function
1	SSR2 to load	6	GND
2	SSR1 to GND	7	SSR3 to load
3	SSR2 to GND	8	SSR4 to GND
4	SSR1 to load	9	SSR3 to GND
5	GND	10	SSR4 to load

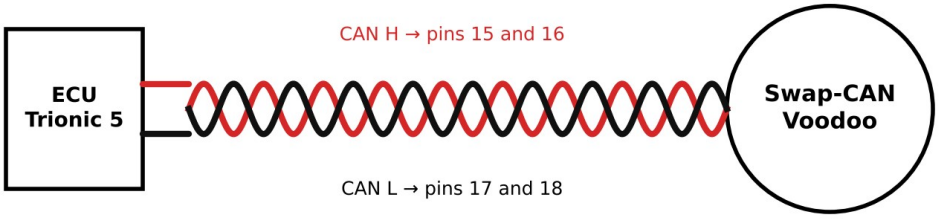
Power supply

It is very important to use **all** the positive supply pins (1, 2 and 4) and **all** the ground pins (22, 23 and 24). The wires in the 24-pin and 10-pin cables are 28 AWG, and sharing the current between them keeps them from heating up. The positive must be a **+15** feed (DIN terminal 15: +12 V that is only live with the ignition on).



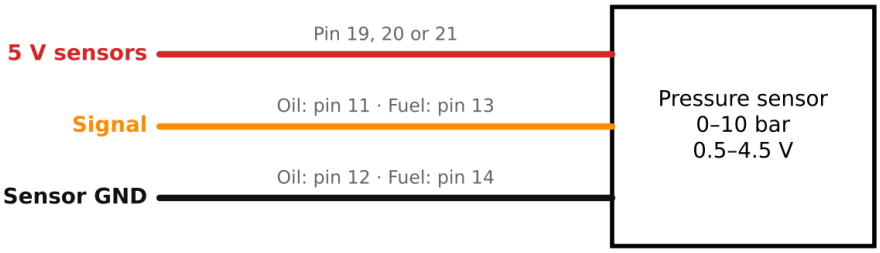
CAN bus

It is very important that the **CAN H** and **CAN L** lines are **twisted together** all the way to the ECU, to minimise transmission errors. The connector has two pins for each line (15–16 CAN H, 17–18 CAN L).



Pressure sensors (oil and fuel)

The oil pressure and fuel pressure sensors are connected in the same way. They are **3-wire** sensors (5 V supply, signal and ground). The colours in the diagram are only a guide.

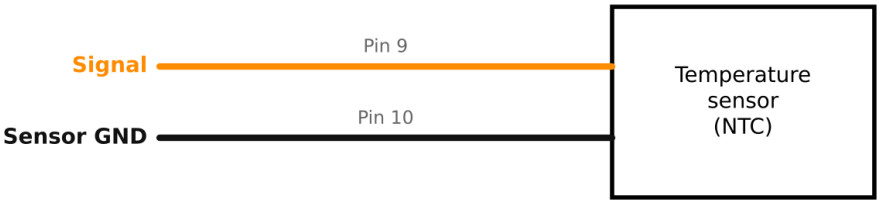


The sensors may or may not be included in the package. If you want to buy them separately, the gauge is calibrated for a sensor with these characteristics:

Feature	Value
Measuring range	0–10 bar (gauge pressure)
Supply	5 V
Output	0.5 V at 0 bar · 4.5 V at 10 bar (linear)
Connection	3 wires: 5 V, signal, ground

Oil temperature sensor

The oil temperature sensor uses only **two wires**, and their order does not matter, because it is simply a resistor whose value changes with temperature (NTC). The colours in the diagram are only a guide.



If you buy the sensor separately, it must have this resistance curve (it is the one the gauge uses to calculate the temperature):

Temperature	Resistance	Temperature	Resistance
20 °C	62,320 Ω	90 °C	4,430 Ω
30 °C	40,340 Ω	100 °C	3,320 Ω
40 °C	26,650 Ω	110 °C	2,550 Ω
50 °C	17,950 Ω	120 °C	1,970 Ω
60 °C	12,320 Ω	130 °C	1,510 Ω
70 °C	8,610 Ω	140 °C	1,170 Ω
80 °C	6,140 Ω	150 °C	900 Ω

Oil temperature is shown from 31 °C upwards; below that a "-" is shown. Above 149 °C, "150+" is shown.

AEM wideband sensor

The wideband AFR is **not** connected to the gauge: the analog output of the AEM wideband controller is connected to **pin 69 of the Trionic 5 ECU**, and the Swap-CAN Voodoo reads it through the ECU. Connect the wideband's power and ground as described in its own manual.



The gauge converts the voltage into AFR with the AEM formula: **AFR = 2.375 × V + 7.3125** (petrol). Any other wideband with a 0–5 V output that uses this same conversion is also compatible:

Voltage	AFR	Lambda	Voltage	AFR	Lambda
0.0 V	7.31	0.50	3.0 V	14.44	0.98
0.5 V	8.50	0.58	3.5 V	15.63	1.06
1.0 V	9.69	0.66	4.0 V	16.81	1.14
1.5 V	10.88	0.74	4.5 V	18.00	1.22
2.0 V	12.06	0.82	5.0 V	19.19	1.31
2.5 V	13.25	0.90			

Lambda calculated as AFR / 14.7. On screen, AFR is only shown between 8.55 and 17.95; outside that range a "-" is shown.

Side/parking lights (+58)

If you want the screen brightness to adjust automatically, connect **pin 7** to the positive feed of the **side/parking lights (+58)**, i.e. DIN 72552 terminal 58, which is live whenever the side lights or headlights are switched on. With the lights on, the brightness drops to **60 %** of the value set in Settings; when they are switched off it returns to the set value. This helps to avoid glare at dusk and unnecessary handling of the display while driving.



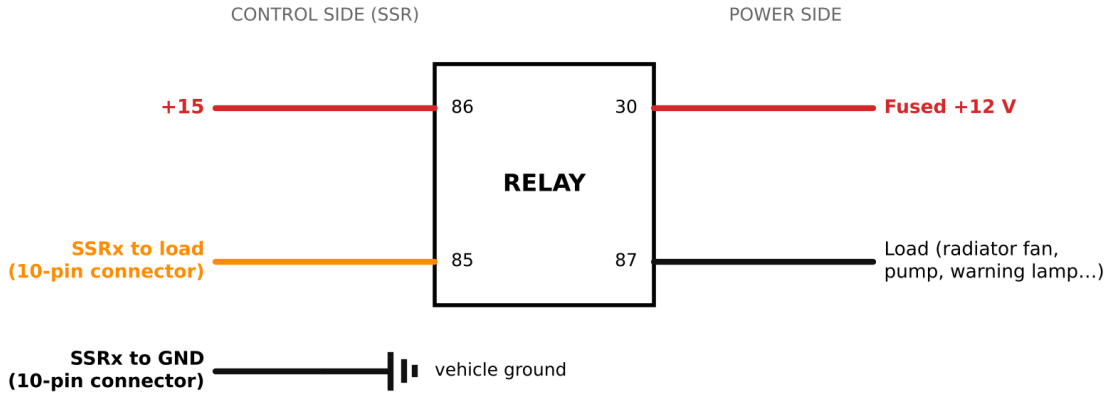
SSR outputs

An **SSR** (*Solid State Relay*) is an electronic switch with no moving parts. The gauge opens and closes it according to the condition you choose on the alarms screen. The gauge has **four** SSRs (SSR1 to SSR4), and each one can switch up to **400 mA**.

Each SSR has two terminals on the 10-pin connector:

- **SSRx to load (LOAD):** connects to the load you want to control, normally the coil of a relay (terminal 85).
- **SSRx to GND (LOAD GND):** connects to a **vehicle ground**, not to an internal ground of the gauge itself.

When the condition is met, the SSR connects its two terminals and the load is grounded. In other words, it works as a ground-side (low-side) switch.



Example: SSR controlling a radiator fan relay.

Since 400 mA is not enough for a motor, the SSR is normally used to switch a **relay**: relay terminal 86 to a **+15** feed, terminal 85 to "SSRx to load", and "SSRx to GND" to a vehicle ground. The relay then switches the power load (terminals 30 and 87). Typical uses:

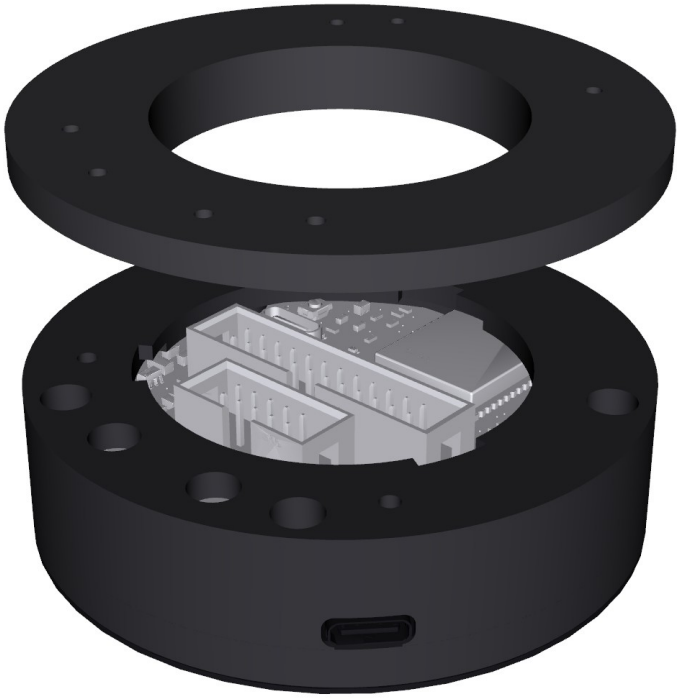
- Switching on the radiator fans according to coolant temperature.
- Switching on an intercooler water pump according to intake air temperature or boost pressure.
- Switching on a warning lamp or an external buzzer on low oil pressure or knock.
- Switching any accessory according to RPM, speed, gear, power or torque (see the list of conditions in the Alarms and SSR chapter).

For safety, if the board that controls the SSRs stops receiving commands from the screen for more than 2 seconds, it switches all four SSRs off.

PWM output: technically, the SSRs can generate a PWM signal (for example, for a fan speed controller). This function is **not yet included in the firmware**; at present the SSRs only work as on/off switches.

Mounting in the vehicle

The gauge is mounted on a **fixed base** that stays bolted in the car. The gauge locks onto the base with a **bayonet fitting** (a twist of about 12°) and three **magnets**. This means it can be removed and refitted in seconds, and one gauge can be used in several cars with Trionic 5: you just need a base installed in each of them.



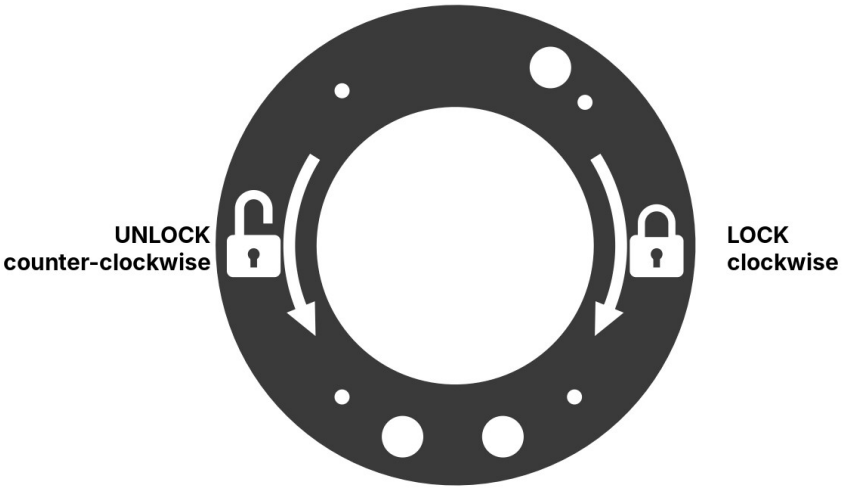
Gauge and mounting base.

Installing the base

- The base is fixed to the vehicle through its **4 holes for threaded inserts**.
- The 24-pin and 10-pin cables go through the central hole of the base to the gauge connectors.
- Leave enough cable to be able to unplug the cables with the gauge taken out of the base.

Fitting and removing the gauge

The face of the base has two engraved arrows with a padlock:



Face of the base, seen from the front.

1. Connect the 24-pin and 10-pin cables to the gauge **with the ignition off**.
2. Place the gauge on the base turned about 12° counter-clockwise (looking at the screen from the front), so that the three lugs of the base go into their slots.
3. Push until it seats and **turn it clockwise** up to the stop (padlock closed). The magnets hold it in that position.
4. To remove it, turn it **counter-clockwise** (padlock open) and pull it out.

[PENDING: add a photo of the gauge mounted in the vehicle]

First start-up

When you switch the ignition on, the gauge shows the loading screen with messages that tell you which step it is at:

Message	Meaning
Attempting handshake…	The screen is establishing communication with the board that reads the ECU.
Handshake not possible!	The screen gets no reply from the CAN board. It keeps retrying on its own; see Troubleshooting.
Different versions! Screen x / CAN y	The two parts of the gauge have different firmware versions. Update both to the same version.
Downloading symboltable from the ECU…	The ECU symbol table is being downloaded (only the first time, or with a new ECU).
Loading…	Normal data loading.
No CAN communication!	The ECU does not answer on the CAN bus. The device beeps twice and restarts to try again.

When the CAN board has recognised the ECU and is ready, it gives **two short beeps** and the screen starts showing data.

Symbol table

The first time it is connected to a vehicle, the data display downloads the **symbol table** of the engine control unit (ECU). This can take between 20 and 40 seconds. The table is stored, so it does not need to be downloaded again at later start-ups as long as the same ECU is used.

Not all ECU software versions are the same or have the same symbol table. For this reason, some data will not be available with certain software versions; this is not a fault of the Swap-CAN Voodoo.

If the data display is switched off in the middle of a CAN communication frame, unread information may be left on the bus. When it is connected again, the display may detect the ECU as "unknown" and download the symbol table again. This causes no problem other than the waiting time; operation is not affected.

An example of the content of this table:

...10D80002Mis200_limit
10DA0002Mis1000_limit...

Here we see information about the *misfire* limits for 200 and 1000 combustions. The first four digits give the address of that value in the SRAM (for Mis200_limit, 0x10D8). The next four give the length of the value in the SRAM (in this case, 2 bytes). Finally comes the name of the symbol inside the ECU, in this case Mis200_limit. The symbol table has roughly 700 to 950 symbols, hence the reading time. If the ECU answers with too few symbols, it is assumed that there was a transmission error or that the table is damaged, and the device restarts to begin again.

Using the gauge

Navigating between screens

You move between screens by **swiping**: swipe left to go to the next screen and right to go back to the previous one. The sequence is circular: after the last screen you return to the first one. When loading has finished, the gauge shows the main screen. The order is:

1. **Main:** the most important data while driving.
2. **Data 2:** oil, fuel, battery and misfires.
3. **Boost:** everything related to boost pressure control.
4. **Runtime:** speed, gear, engine speed, throttle and load.
5. **Sensors:** the readings of the gauge's own sensors.
6. **AFR:** mixture, fuel corrections and knock.
7. **Miscel.:** ECU status indicators and errors.
8. **Alarms & SSR:** alarm and output settings (see the Alarms and SSR chapter).
9. **Settings:** brightness, MAP sensor, logs and tables (see Settings, below).

Each screen is described in detail in the **Screens and data** chapter.

Settings

- **Brightness:** horizontal slider to adjust the screen brightness (0–100).

- **MAP sensor type:** select the MAP sensor fitted to the engine: 2.5, 3.0, 3.5, 4.0 or 5.0 bar. **If it does not match the actual sensor, the scaling of boost pressure, target pressure and the diagnostic tables will be wrong.**
- **Start / Stop log:** starts and stops log recording (see the Logs chapter).
- **AFR FEEDBACK RESET:** press and hold to clear the AFR feedback table.
- **Knock map reset:** press and hold to clear the gauge's own knock map.

Settings are stored and kept when the gauge is switched off.

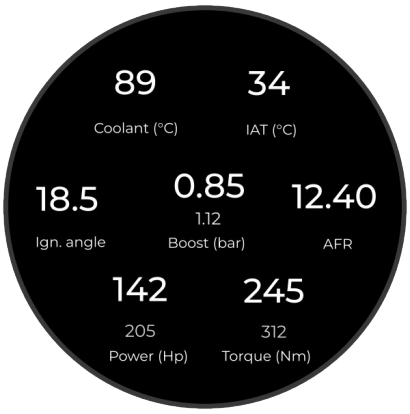
Screens and data

This chapter explains, screen by screen, what each value is, where it comes from and how to interpret it. The pictures show each screen with **example values**; they are not a real measurement.

A dash (-) instead of a number means the value is not valid at that moment: the sensor is not connected, it is outside its measuring range, or the value has not yet been received from the ECU.

Most of the data is calculated by the **Trionic 5 ECU**; the gauge only reads and displays it. Values measured by the gauge's own sensors (oil pressure and temperature, fuel pressure) or by the AEM wideband are pointed out in each case. Some values are shown **raw**, exactly as the ECU stores them, without converting them to a unit: they are useful for comparing one situation with another rather than for their absolute value.

Main



Main screen (example values).

This is the everyday screen: at a glance it shows temperatures, boost, mixture, ignition and performance.

Value	What it is	How to interpret it
Coolant (°C)	Engine coolant temperature, measured by the ECU.	With a warm engine it is usually between 85 and 95 °C. It changes colour as it approaches the coolant alarm (see Colour warning).
IAT (°C)	Intake Air Temperature, measured by the ECU.	The lower, the better. It rises with boost and in traffic; if it gets very high, the ECU reduces ignition advance to protect the engine. It changes colour as it approaches its alarm.
Boost (bar)	Boost pressure in the intake manifold, in bar **relative** to atmospheric pressure.	It is negative with the throttle closed (vacuum) and positive when the turbo is boosting. It depends on the **MAP sensor** selected in Settings being the right one. It changes colour as it approaches the boost alarm.
Small number under Boost	**Peak boost pressure** reached.	It is held for **25 seconds** after the last peak and then follows the current value again. It lets you read the peak of a pull without taking your eyes off the road for long.
AFR	Air/fuel ratio measured by the **AEM wideband** (if fitted).	Same value as AFR AEM on the AFR screen. On petrol, 14.7 is stoichiometric; lower values are rich and higher values are lean. Shows a dash below 8.6 or above 17.9 (for example, while the sensor is still warming up).
Ign. angle	Ignition timing being applied by the ECU, in degrees of advance before top dead centre.	Includes the ECU's corrections, including knock retard. If it drops suddenly during a pull while boost holds steady, the ECU is retarding the ignition because of knock.
Power (Hp)	Power calculated by the gauge from the ECU torque and RPM.	It is an **indicative and somewhat conservative figure**, not a dyno measurement. The small number below it is the **maximum** reached, held until the gauge is switched off.
Torque (Nm)	Engine torque **estimated by the ECU**.	Also an estimate, useful for comparison. The small number is the **maximum** since the gauge was switched on.

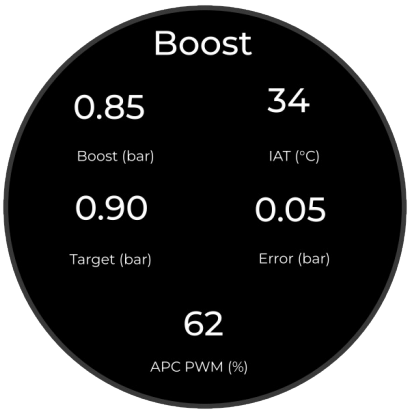
Data 2



Data 2 screen (example values).

Value	What it is	How to interpret it
Oil press. (bar)	Oil pressure, measured by the **gauge's pressure sensor** (optional).	Lower at idle and rising with RPM. Pressure that drops when accelerating, or that is slow to build after starting, indicates a problem: stop the engine and check it. Shows a dash if the sensor is not connected.
Oil temp. (°C)	Oil temperature, measured by the **gauge's temperature sensor** (optional).	Oil takes much longer than coolant to warm up: do not work the engine hard until it reaches operating temperature. Shows a dash below 31 °C and **150+** above 149 °C.
Fuel press. (bar)	Fuel pressure, measured by the **gauge's pressure sensor** (optional).	It should stay stable. If the regulator is referenced to the manifold, fuel pressure rises together with boost. A drop at full load indicates that the pump or filter cannot supply enough flow.
Volt.	Battery voltage, measured by the ECU.	With the engine running, normally between 13.5 and 14.5 V. With the engine off, around 12.5 V. Low values with the engine running indicate a charging fault.
Miss200	ECU **misfire** counter, over windows of 200 engine revolutions. It detects misfires that could damage the catalytic converter.	Normally **0**. If it rises, there are misfires: check the spark plugs, ignition cassette (DI) and mixture.
Miss1000	ECU misfire counter over windows of 1000 revolutions. It detects less severe, emission-related misfires.	Normally 0 or a very low value. A value that keeps rising indicates continuous misfiring.

Boost



Boost screen (example values).

It shows how the ECU's boost pressure control (APC) is working: what it asks for, what it achieves and how hard it is driving the valve.

Value	What it is	How to interpret it
Boost (bar)	Current boost pressure, in relative bar (same as on the main screen).	Compare it with Target to see whether the turbo reaches the requested pressure.
IAT (°C)	Intake air temperature.	Repeated here because the ECU reduces boost and ignition advance when the air is very hot.
Target (bar)	**Target pressure** requested by the ECU at that moment, in relative bar.	Depends on RPM, throttle, gear and the ECU's corrections (for example for knock or temperature).
Error (bar)	Difference between target and actual pressure: **Target − Boost**.	Close to 0: the turbo follows the target well. **Positive:** boost is missing (slow turbo, leak, valve). **Negative:** overboost compared with the request. When accelerating it is normal for it to be positive for a moment while the turbo spools up.
APC PWM (%)	Duty cycle with which the ECU drives the **boost control valve** (APC).	The higher it is, the more boost the ECU is trying to get. If it is very high and boost is still missing (positive Error), look for leaks or a fault in the valve or actuator.

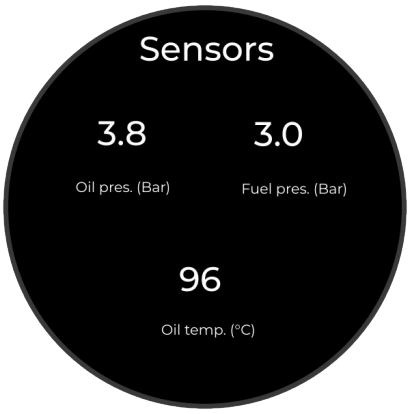
Runtime



Runtime screen (example values).

Value	What it is	How to interpret it
km/h	Vehicle speed, according to the ECU.	It should roughly match the speedometer. If it reads 0 while driving, check the speed signal (see the Speed error on Miscel.).
Gear	Gear engaged, **calculated by the ECU** from speed and RPM.	Shows **N** when the ECU does not detect any gear: in neutral, with the car stopped, or while the speed/RPM ratio does not match any gear (for example, with the clutch pressed). After a gear change it may take a moment to update.
RPM	Engine speed, in revolutions per minute.	Taken directly from the ECU; the same value used by the diagnostic tables.
RPM max.	**Rev limit** (cut-off) set in the ECU map.	Not a maximum reached, but the value programmed in the ECU. Useful to check which limit the loaded map has.
TPS	Throttle position (Throttle Position Sensor), as averaged by the ECU. **Raw value**.	Higher means more throttle. Useful to check that the value rises smoothly as you press the pedal and always reaches the same value at full throttle.
Max. speed	**Speed limit** set in the ECU map, in km/h.	Like RPM max., it is a programmed value, not a maximum reached.
Load	Average engine load calculated by the ECU. **Raw value**.	Higher means more load (more air per cycle). Useful for comparing situations with each other.

Sensors



Sensors screen (example values).

It shows in large digits the readings of the **gauge's own sensors**: oil pressure, fuel pressure and oil temperature. They are the same values as on the Data 2 screen and are interpreted in the same way (see above). If a sensor is not fitted, its value shows a dash.

AFR



AFR screen (example values).

It shows the air/fuel mixture, the corrections the ECU applies to fuelling and knock detection. Fuel corrections are at the top, the oxygen sensors in the middle and knock at the bottom.

Mixture and fuel corrections

Value	What it is	How to interpret it
Lambdaint	**Lambda integrator:** short-term fuel correction applied by the ECU in closed loop, based on what its own lambda sensor reads.	With the ECU in closed loop it moves up and down continuously: the ECU keeps adjusting the mixture to hold it at lambda 1. If it sticks at one end, the ECU is correcting as much as it can: check the fuel map, injectors or lambda sensor. Outside closed loop (full load, cold engine) it does not change.
Long fuel adapt.	**Long-term fuel adaptation:** factor the ECU has learned over time, which multiplies the whole fuel map.	**1.000** means no correction. Above 1 (for example 1.050): the ECU is adding 5 % fuel because the engine tends to run lean. Below 1: it is removing fuel. The further from 1, the further the map is from what the engine actually needs. Updated every 20 seconds.

Oxygen sensors

Value	What it is	How to interpret it
AFR AEM	Air/fuel ratio measured by the **AEM wideband**, connected to the ECU (pin 69).	This is the **reliable** mixture reading across the whole range, including full load. On petrol: 14.7 is stoichiometric, lower is rich and higher is lean. A rich mixture is wanted at full boost; the right value is set by whoever tunes the map. Shows a dash outside the 8.6–17.9 range.
Lambda	**Approximate** lambda value, calculated from the ECU's **original lambda sensor** (narrowband).	This sensor can only tell rich from lean reliably **around lambda 1**. In closed loop it oscillates around 1.00. Values far from 1 are only a rough indication: use AFR AEM to measure the mixture.

Value	What it is	How to interpret it
Lambda ADC	**Raw** reading of the original lambda sensor (0–255, scaled to 0–5 V).	At stoichiometric mixture the sensor gives about 0.45 V (about 23). High values (around 40–45) indicate a rich mixture and low values (around 5) a lean one. In closed loop it should swing between both ends; if it stays fixed, the sensor may be cold or faulty.

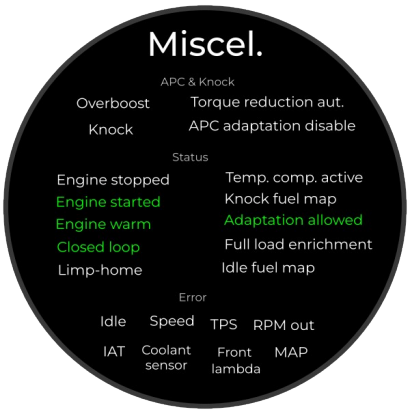
Knock

The Trionic 5 does not use a knock sensor: it detects knock (detonation) by measuring the **ionisation current** across the spark plugs themselves, through the ignition cassette. When it detects knock, it retards the ignition on that cylinder and, if knock persists, it richens the mixture and reduces boost.

Value	What it is	How to interpret it
Knock	Shows whether the ECU **is detecting knock right now**: 1 = yes, 0 = no.	Brief, isolated detections are normal and the ECU corrects them by itself. If it appears often, or always in the same area, check the ignition map, fuel quality and intake air temperature.
Knock (20s)	The same indicator, but it **stays at 1 for 20 seconds** after the last detected knock.	Lets you notice knock even if you were not looking at the screen at that moment.
AD Knock	**Raw** value of the knock signal measured by the ECU.	The higher it is, the stronger the signal. Mainly useful for comparison: a value that clearly rises under certain conditions shows where knock occurs.

To find out in which area of the map knock occurs, use the **Knock Map** and **Knock_Count_ECU** tables on the SD card (see the Diagnostic tables on the SD card chapter).

Miscel.



Miscel. screen (example: engine started, warm, in closed loop and with adaptation allowed).

This screen shows no numbers, only **indicators** that light up according to the ECU status. In **white**, the condition is not active. In **green**, it is active. In **red**, a protection mode or an error is present.

APC & Knock

Indicator	Lights up when...	How to interpret it
Overboost	boost pressure has exceeded the overboost limit in the map.	The ECU cuts fuel injection to protect the engine. If it appears, check the boost control (APC valve, actuator, hoses).
Torque reduction aut.	the ECU is reducing torque on its own.	It is an automatic ECU protection or limitation; if it appears often for no obvious reason, check the other indicators and the logs.
Knock	the ECU is detecting knock.	Same as the Knock value on the AFR screen.
APC adaptation disable	boost control adaptation is disabled.	The ECU is not learning boost corrections at that moment. This is normal under some operating conditions.

Status

Indicator	Lights up when...	How to interpret it
Engine stopped	the engine is stopped.	Normal with the ignition on and the engine not started.
Engine started	the engine is running.	Normal with the engine running.
Engine warm	the engine has reached operating temperature.	Until it lights up, the ECU applies cold-engine corrections.
Limp-home	the ECU has entered **limp-home mode** (red).	The ECU limits operation because of a serious fault. Check the errors at the bottom and read the ECU fault codes.
Closed loop	the ECU is working in **closed loop**: it adjusts the mixture with the lambda sensor.	Normal at part load with a warm engine. It is not active at full load or with a cold engine.
Temp. comp. active	temperature fuel compensation is active.	Common when the engine or intake air is outside its normal temperature.
Knock fuel map	the ECU is using the **knock fuel map** (red).	When knock persists, the ECU richens the mixture to protect the engine. If it appears, there is sustained knock: check the map, fuel and temperatures.

Indicator	Lights up when...	How to interpret it
Adaptation allowed	the conditions are met for the ECU to **learn** fuel adaptation.	This is when the ECU updates Long fuel adapt.
Full load enrichment	**full-load enrichment** is active.	Normal at full throttle: the ECU leaves closed loop and richens the mixture.
Idle fuel map	the ECU is using the **idle fuel map**.	Normal with the engine idling.

Error

Errors light up in **red** when the ECU detects a fault in that system. If any of them lights up, read the ECU fault codes for details.

Indicator	Fault detected in...
Idle	the idle control.
Speed	the vehicle speed signal.
TPS	the throttle position sensor.
RPM out	the ECU's RPM signal output.
IAT	the intake air temperature sensor.
Coolant sensor	the coolant temperature sensor.
Front lambda	the original lambda sensor.
MAP	the manifold pressure (MAP) sensor: the reading is not plausible, for example when the ignition is switched on.

Alarms and SSR

On the **Alarms & SSR** screen you choose, for each value, the level at which the alarm sounds. While an alarm is active, the gauge gives **short, repeated beeps**. All options with a numeric value include a **TEST** option, which keeps the alarm on permanently so you can check that the buzzer works.

Alarm	Triggers when...	Options	Default
Coolant	coolant temperature is equal to or higher than the value	90, 95, 99, 103, 107, 112, 120, 130 °C, TEST	103 °C
IAT	intake air temperature is equal to or higher than the value	30 to 80 °C in steps of 5, TEST	60 °C
Boost	boost pressure is equal to or higher than the value	0.6, 0.8, 1 to 2.5 bar in steps of 0.1, TEST	1.8 bar
Oil P	oil pressure is equal to or **lower** than the value, with the engine above 500 rpm	No, 0 to 1 bar in steps of 0.1, TEST	No
Oil T	oil temperature is equal to or higher than the value	No, 110 to 155 °C in steps of 5, TEST	No
Knock	Yes: with any knock. >3s: only with sustained knock (see below). No: disabled.	Yes, No, >3s	>3s

Colour warning

Before the alarm is reached, the value on screen **changes colour** to warn you that it is getting close: it goes from **white** to **yellow** and then to **red**, which it reaches exactly at the alarm value. The colour change starts:

- **Coolant:** 6 °C before the alarm value.
- **Intake air temperature (IAT):** 10 °C before.
- **Boost pressure:** 0.6 bar before.

For example, with the coolant alarm set to 103 °C, the value is white up to 97 °C, starts turning yellow from 98 °C, and at 103 °C it is red and the alarm sounds.

Sustained knock alarm (>3s)

Occasional knock that the ECU corrects by itself is normal and is recorded in the logs. With the **>3s** option, the alarm only sounds when there is knock **continuously for at least 3 seconds** (it counts as the same burst as long as no more than 1 second passes without knock). This avoids warnings for isolated detections.

SSR output conditions

On the same screen you choose the condition for each SSR (SSR1 to SSR4). The SSR is switched on while the condition is met and switched off as soon as it is no longer met:

Group	Available conditions
—	Nothing (SSR always off)
Coolant	Coolant > 90, 93, 95, 98, 100, 103, 106 °C
Intake air	IAT > 25, 30, 35, 40, 45, 50, 55, 60 °C
Oil	Oil P < 0.2, 0.4 bar · Oil T > 106, 110, 115, 120, 125, 130, 135 °C
Knock	Knock (while the ECU detects knock)
Boost	Boost > 0.5, 0.8, 1, 1.2, 1.4, 1.6, 1.8 bar
Speed	Kmph > 60, 100, 120, 150, 180
RPM	RPM < 1000, 2000, 3000, 4000, 5000 · RPM > 2000, 3000, 4000, 5000, 6000
Power and torque	HP > 100, 200, 300, 400, 500 · Nm > 100, 200, 300, 400, 500, 600
Gear	Gear = 1, 2, 3, 4, 5, N
Fuel	Fuel P < 2, 3, 4 bar · Fuel P > 3.5 bar
Test	TEST ON (SSR always on, to check the installation)

The condition has no hysteresis: if the value hovers right around the threshold, the SSR will switch on and off just as quickly.

Logs

Logs are data recordings made by the display. They are stored on a **microSD card** that can be inserted without removing the device: the slot is at the top left of the dial. The recording rate is approximately **20 Hz**.

Recording a log

1. Make sure a microSD card is inserted and switch the display on.
2. Go to the **Settings** screen and press the **Start** button (Start a log).
3. Before starting, the card size and free space are shown (for example, "SD: 25000 / 30000 MB"). Make sure the card is not full.
4. While recording, the button changes to a red **Stop** button and the name of the file being recorded is shown (for example, "File: A5IZV1NL.36L_3.t5l").

5. To finish, press **Stop**. "Closing..." appears, followed by "Log saved".

Log files

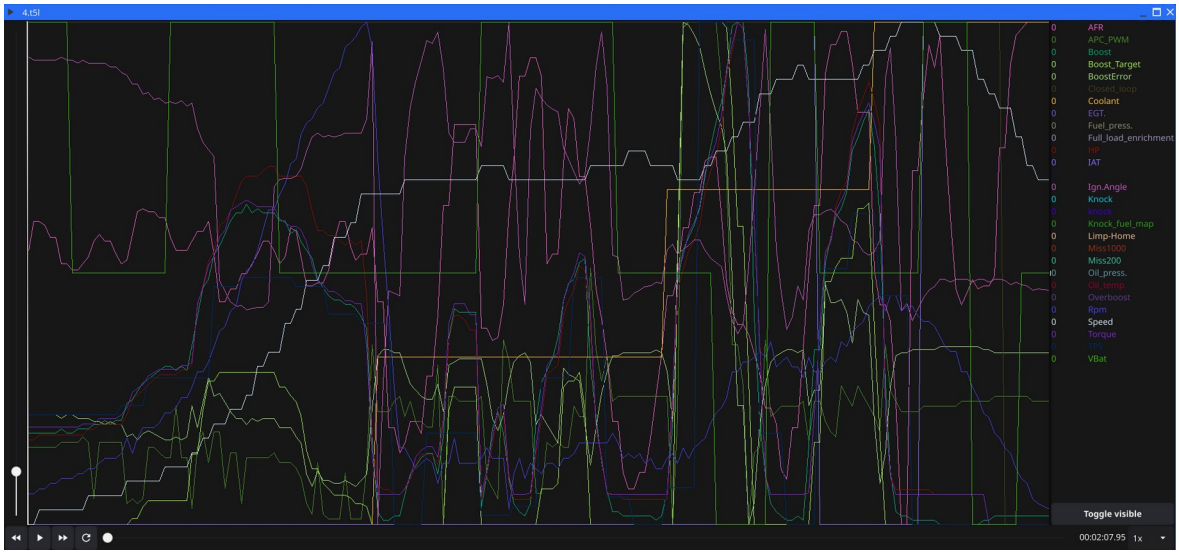
- Name: **<ECU software version>_<number>.t5l**, for example **A5IZV1NL.36L_1.t5l**. This makes it easy to tell which car each log comes from.
- They are numbered from 1 to 299. If all 299 already exist, "No free files!" appears: copy the logs to your computer and delete them from the card.
- The time of each line is the time elapsed since the gauge was switched on (the date shown is fixed, because the gauge has no real-time clock).
- Recorded data: RPM, boost pressure, ignition angle, coolant, IAT, AFR, TPS, target pressure and error, APC PWM, knock, misfire (total and per cylinder, 200 and 1000), speed, oil pressure and temperature, fuel pressure, battery voltage, status flags (knock fuel map, overboost, limp-home, closed loop, idle, full-load enrichment), power, lambda ADC, gear and torque.

Analysing the logs

Once the log has been recorded and the session is over, it is time to analyse it. Two tools developed by the Saab community are available for this:

- **TXLogger**
- **T5Suite**

With the first one, once downloaded from txlogger.com and installed, double-click the file you want to analyse and a chart with all the recorded data will open. A legend on the right helps you interpret them and lets you hide the ones that are not relevant.



Log opened in TXLogger.

Diagnostic tables on the SD card

In addition to the logs, the gauge automatically creates some tables on the microSD card that help you review the ECU map. They are created without having to record a log, as long as a card is inserted.

AFR Feedback

File **AFR_Feedback_<ECU>_<MAP sensor>.csv** (for example, AFR_Feedback_A5IZV1NL.36L_3bar.csv). It uses the same axes as the ECU **fuel map**: pressure in columns (relative bar) and RPM in rows.

- **First table:** average AFR measured by the wideband in each cell of the fuel map.
- **Samples per cell:** number of samples in each cell. The more samples, the more reliable the average.
- Data is only recorded with the engine above 800 rpm, coolant at 70 ºC or more and a valid AFR (between 8 and 22).

Knock Map

File **Knock_Map_<ECU>_<MAP sensor>.csv**. It uses the same axes as the ECU's main **ignition map** (Ign_map_0), so each cell with knock points to the exact ignition map cell that should be reviewed. It contains three tables:

- **Knock rate:** knocks per 100 samples in the cell. It lets you compare areas where different amounts of time were spent.
- **Knock events:** number of times knock started in that cell.
- **Samples per cell:** number of readings in each cell (how long the car was driven in that area).

Knock_Count_ECU

File **Knock_Count_ECU_<ECU>_<MAP sensor>.csv**. It is a copy of the knock counters kept by **the ECU itself** in its memory: the knock map per cell (with the same axes as the ignition map) and the total knock count per cylinder (Cyl1 to Cyl4). The gauge reads it at start-up and then every 5 minutes (the per-cylinder counters every minute). Just switching the gauge on is enough to copy to the card what the ECU has accumulated.

Operation and clearing

- The **AFR Feedback** and **Knock Map** tables accumulate between sessions: each time you switch the gauge on, new data is added to the existing data. They are saved to the card every 30 seconds if there is new data.
- Next to each .csv there is a **.dat** file with the internal data. **Do not delete it** if you want to keep accumulating data.
- There is a separate file for each ECU software and each MAP sensor. If the map axes change in the ECU, the previous table is discarded automatically.
- To start from scratch, press and hold the corresponding reset button in **Settings**. "AFR table reset" or "Knock map reset" will appear.

Opening the tables

The .csv files open directly in **Microsoft Excel** or **LibreOffice Calc**. They use a semicolon as separator and a comma as decimal mark, so the columns appear already separated with Spanish/European regional settings. If your system uses a decimal point, choose the semicolon as separator and the comma as decimal separator when importing. To see them with colours, apply **conditional formatting → colour scale** to the table in Excel. In the future they will also be viewable with the **Voodoo Updater** program.

How it works

The Swap-CAN Voodoo data display reads the SRAM memory of the Trionic 5 engine control unit (ECU), which uses a Saab-proprietary CAN bus protocol.

Update rate

Not all data is updated at the same rate: the ECU has to process every request it receives over CAN, and asking for everything all the time would waste resources. For this reason, the data is split into groups by update rate:

Group	Every	Examples
Very slow	20 s	Long-term fuel adaptation (adapt_injfaktor).
Slow	1 s	Gear engaged, intake air and coolant temperatures, battery voltage, misfire errors.
Medium	300 ms	Diagnostic symbols, engine load and misfire markers.
Semi-fast	100 ms	Vehicle speed and throttle position (TPS).
Fast	40 ms	Intake pressure, RPM, AFR, APC PWM, knock, torque, ignition angle and program status.

Data is sent to the screen and to the log every 50 ms, i.e. about 20 times per second.

Display and brightness

The display has a resolution of 480 × 480 pixels, is a touchscreen and is backlit. The brightness can be adjusted in two ways:

- From **Settings**, by moving the horizontal slider (Brightness).
- Automatically, if the positive feed of the **side/parking lights (+58)** has been connected to pin 7: with the lights on, the brightness drops to 60 % of the set value, and when they are switched off it returns to the set value.

Updating

Before updating the device, make sure the software version is compatible with your hardware. The data display consists of **two sub-modules** that are updated independently, and **both must run the same version**; otherwise the loading screen will show "Different versions!".

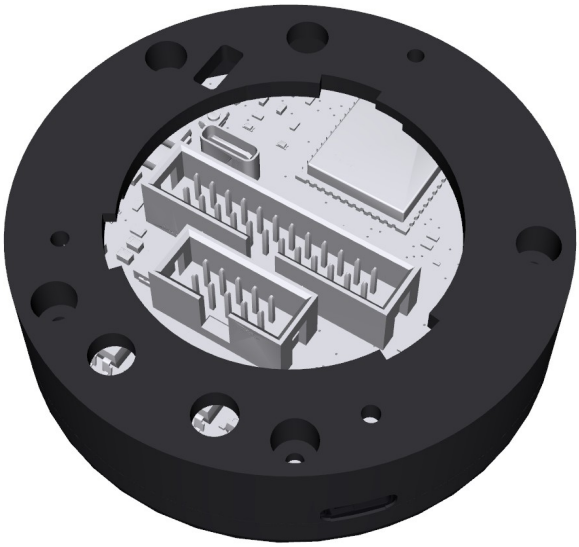
USB-C ports

The **display** sub-module is updated through the USB-C port located on the **right** of the screen, looking at it from the front. Only use this port to update the display.

Do not use the USB-C port on the left: it shares the communication lines with the CAN board. With a cable plugged into it, the Voodoo Updater does not detect the display, and the display may lose communication with the CAN board.



The sub-module that **communicates with the ECU** over CAN is updated through the USB-C port on the back of the device. It may have a plastic cap to keep dust out:

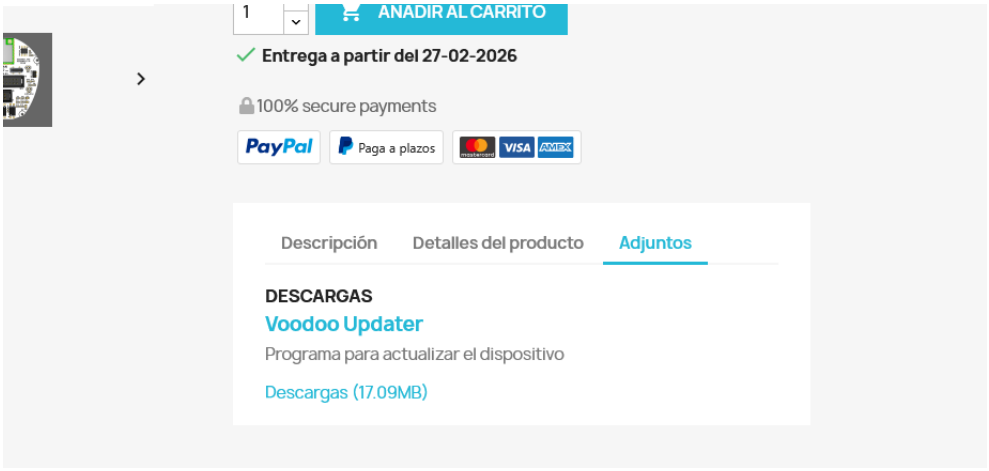


Back of the gauge: USB-C port of the CAN board next to the connectors.

Very important: do not load the display firmware onto the sub-module that communicates with the ECU, or vice versa.

Voodoo Updater

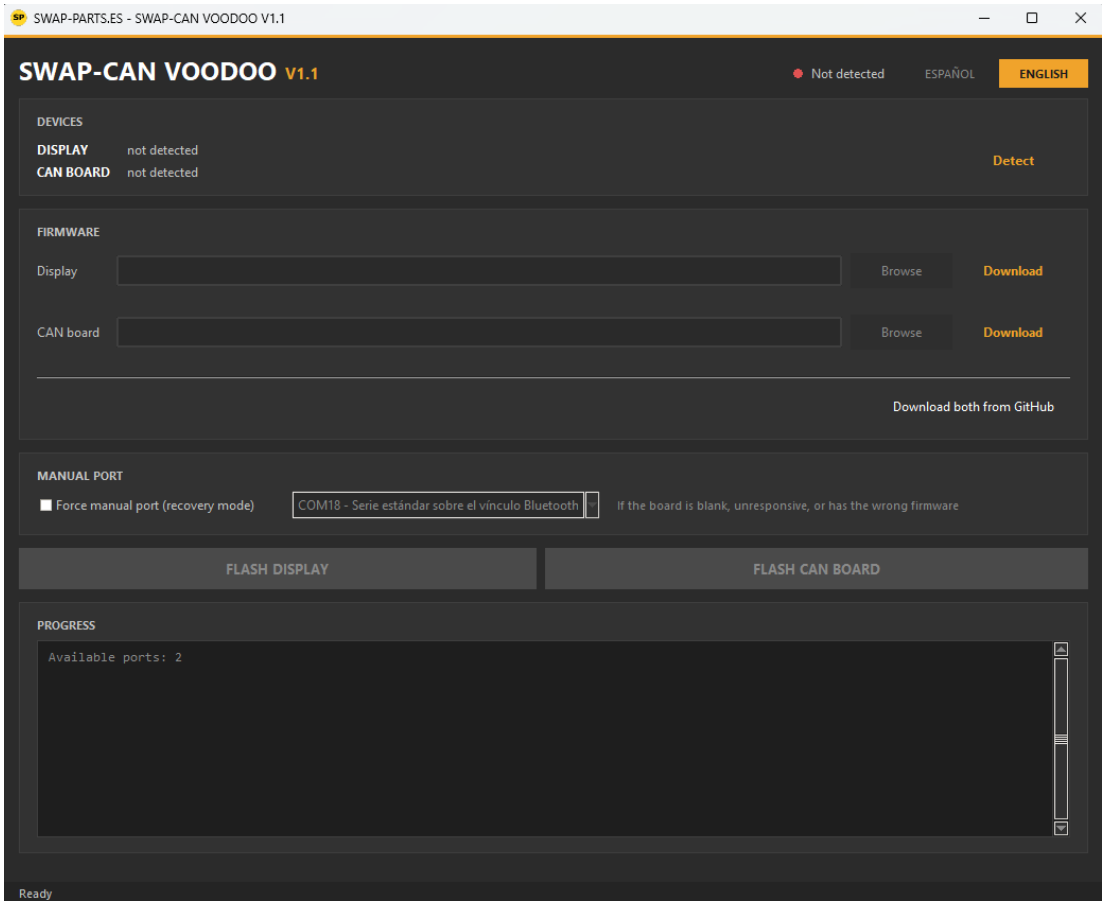
The official update tool, **Voodoo Updater**, can be downloaded from the product page on swap-parts.es, in the **Adjuntos** (Attachments) tab:



How it works

At start-up, each board of the gauge sends an identification message over USB with its type (display or CAN board) and its firmware version. When you press **Detect**, the Voodoo Updater scans the USB ports, recognises each board by that message and only allows each board to be flashed with the firmware that belongs to it. The firmware files are downloaded from our public GitHub repository, which always contains the latest published version.

- **Operating system:** Windows (tested on Windows 11).
- **Internet connection:** only needed to download the firmware; a .bin file saved on the computer can also be used.
- **USB-C data cable:** some cables only charge and do not carry data; if the gauge does not appear, try another cable.
- **Language:** the program is available in Spanish and English; switch with the ESPAÑOL / ENGLISH buttons at the top right.



Voodoo Updater main window, with no board connected.

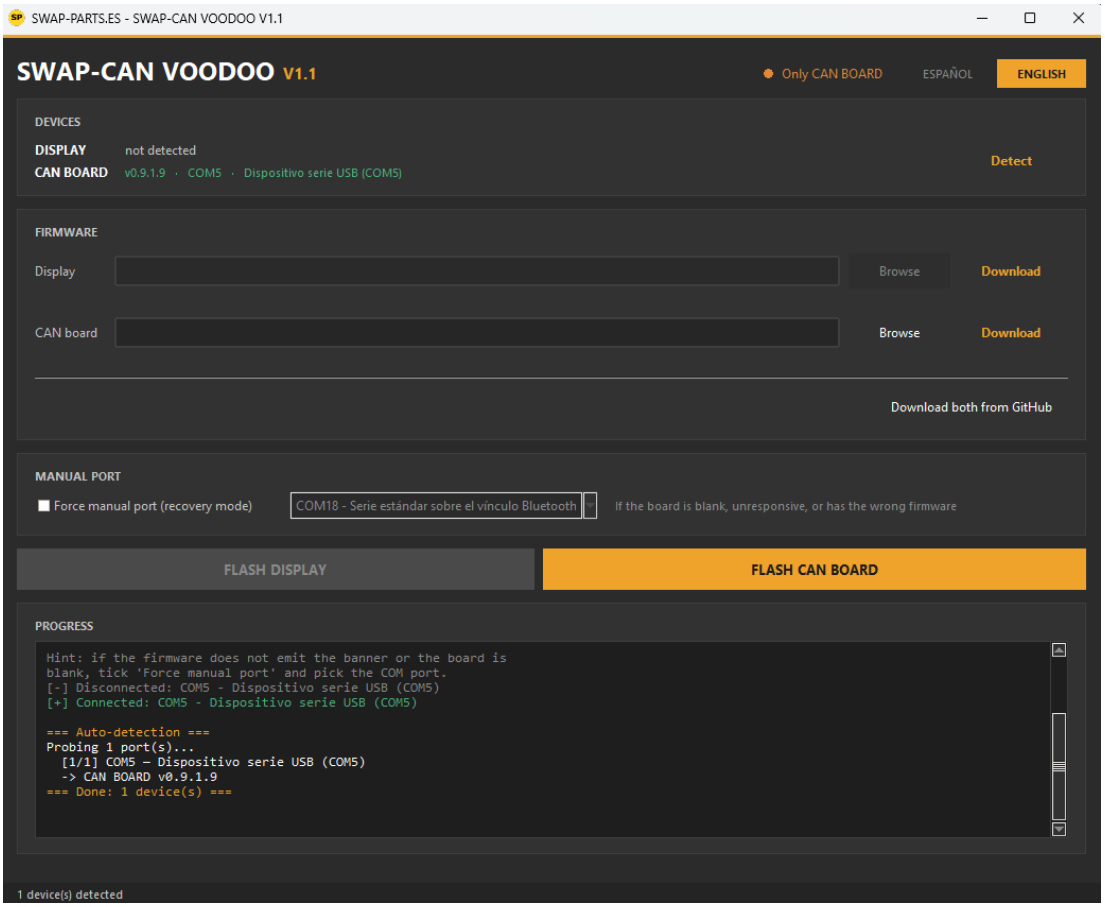
The window has four areas:

- **Devices:** shows whether the display and the CAN board have been detected, with their installed version and COM port. The **Detect** button searches for connected boards; press it every time you plug in a board.
- **Firmware:** the file to be installed on each board. **Download** gets the latest version from GitHub; **Browse** lets you choose a .bin file on your computer; **Download both from GitHub** downloads both at once.
- **Manual port:** recovery mode (see below).
- **Progress:** log of everything the program does.

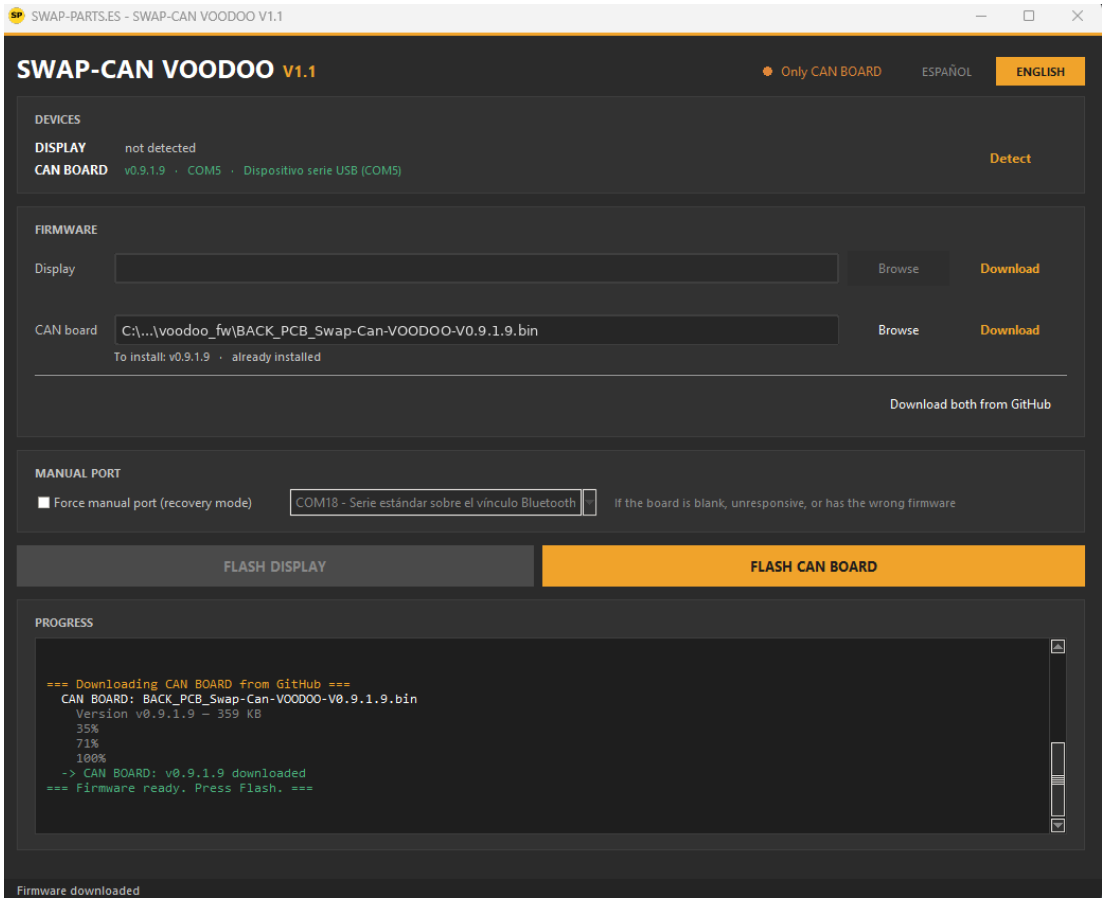
Automatic update (recommended)

1. Switch the vehicle ignition off, or remove the gauge, so that the board you are going to update **starts up when you plug in the USB**.
2. Open the Voodoo Updater and connect the USB-C of the board you want to update: the one on the **right of the screen** (looking at the screen from the front) for the display, or the one on the **back** for the CAN board.
3. Press **Detect**. The board appears under **Devices** with its version and port; for example, "CAN BOARD v0.9.1.9 · COM5". If it does not appear, unplug and reconnect the USB and press **Detect** again.
4. Press **Download** next to that board. Progress shows the download and, below the file field, the version to be installed compared with the installed one: "update available", "already installed" or "this would be a downgrade".

5. Press **FLASH DISPLAY** or **FLASH CAN BOARD**, check the details in the confirmation window and accept.
6. **Do not unplug the USB** until the message saying the board has been updated successfully appears. The board restarts by itself when finished.
7. Press **Detect** to confirm it now has the new version.
8. Repeat the process with the other board. **Both must end up with the same version.**



CAN board detected after pressing Detect (v0.9.1.9 on COM5).



Firmware downloaded and ready to flash.

Manual update

If you already have the firmware file on your computer, for example because we sent it to you, press **Browse** next to the corresponding board and select it. The rest of the process is the same. The files are merged binaries (.bin) and their name includes the board type and the version; for example, **BACK_PCB_Swap-Can-VOODOO-V0.9.1.9.bin** for the CAN board.

Recovery mode (manual port)

If a board is blank, does not respond or has the wrong firmware, it will not send its identification and will not be detected. In that case:

1. Tick **Force manual port (recovery mode)** and choose the board's COM port from the list (the one that appears when you connect it).
2. Choose the **correct** firmware file for that board and press the corresponding flash button.

Warning: in manual mode the program **cannot check** which board is connected. Make sure the firmware matches the board: the display firmware on the display and the CAN board firmware on the CAN board.

Possible problems

Problem or message	Solution
The board is not detected ("not detected")	The identification is only sent when the board starts up. Unplug and reconnect the USB with the ignition off and press **Detect**. Sometimes closing and reopening the program solves it. If it is still not detected, use recovery mode.

Problem or message	Solution
BUSY or unreachable	Another program is using that port (for example, a serial monitor). Close it and press Detect again.
Wrong port	The selected port belongs to the other board. Connect the correct USB.
Missing firmware / File not found	Press Download or Browse to choose the .bin file.
No connection	The computer has no internet access. Connect it or use a previously downloaded file.
GitHub rate limit reached	Too many downloads in a row. Wait a few minutes.
Flashing failed	Check that the USB cable is a data cable and that the port is correct. If the board does not enter programming mode, hold BOOT, press RESET, and try again.

Care and storage

No special care is required when the device is not fitted in the vehicle: just keep it in a dry place where it cannot get wet or be splashed by liquids, like any other electronic device.

Inside the vehicle, the temperature must not exceed 50 °C or drop below 0 °C. Strong sunlight on the screen surface is not recommended, as it may damage it. The same storage principles apply as outside the vehicle.

Messages and troubleshooting

Loading screen messages

Message	What it means	What to do
Attempting handshake…	The screen is looking for the CAN board.	Wait. If it does not change, check the gauge's power supply.
Handshake not possible!	The CAN board does not answer the screen.	Check the power supply and all the grounds. The gauge keeps retrying on its own.
Different versions! Screen x / CAN y	Different firmware on each sub-module.	Update both to the same version.
Downloading symboltable from the ECU…	Symbol table download.	Wait 20–40 s; it only happens the first time.
Loading…	Normal start-up.	—
No CAN communication!	The ECU does not respond.	Check CAN H / CAN L all the way to the ECU, and that the ECU is powered.

Settings messages (SD and tables)

Message	What it means
SD: free / total MB	Free and total card space, when starting a log.
Opening file…	Creating the log file.
File: name.t5l	Name of the file being recorded.
Closing…	Closing the log after pressing Stop.
Log saved	Log saved correctly.
SD: Not detected	No card, or the card is not detected. Insert it or try another card.

Message	What it means
SD Error	The log could not be started on the card.
SD: Write error!	Error writing to the card (full, damaged or removed during recording).
No free files!	There are already 299 logs on the card. Copy and delete the old ones.
AFR table reset	AFR feedback table cleared.
Knock map reset	Knock map cleared (also shown if the map axes change in the ECU).

Audible signals

- **Two short beeps at start-up:** the CAN board has recognised the ECU and starts reading data.
- **Two long beeps and restart:** the ECU has not answered over CAN.
- **Short repeated beeps:** an alarm is active (see Alarms and SSR).

Common problems

Symptom	Probable cause / solution
No connection with the ECU	Check the CAN wiring from the gauge to the ECU; CAN H and CAN L twisted together and not swapped.
The gauge restarts	Check the power supply (all +15 and all GND pins) and the wiring between the gauge and the ECU. The screen restarts by itself if it receives no data from the CAN board for 5 seconds.
Boost pressure looks wrong	Select in Settings the MAP sensor fitted to the engine.
Oil pressure or temperature shows "-"	Sensor not connected, out of range or wrongly wired (signal and sensor GND). Oil temperature shows "-" below 31 °C.
AFR shows "-"	The wideband is not connected to ECU pin 69, is still warming up, or its reading is outside 8.55–17.95.
Some data is missing	The ECU software does not have that symbol, or the symbol table has been removed (see Intended use and compatibility).
Brightness does not drop with the lights on	Check that pin 7 is connected to +58 (side/parking lights, DIN terminal 58).
An SSR does not switch	Check that "SSRx to GND" goes to a vehicle ground and that the selected condition is met. Try TEST ON.

Symbols and terms

Term	Meaning
+15	Ignition-switched positive (DIN 72552 terminal 15): +12 V only with the ignition on.
+30	Permanent battery positive (DIN 72552 terminal 30).
+58	Side/parking lights positive (DIN 72552 terminal 58): live whenever the side lights or headlights are switched on.
GND	Ground (chassis earth).
AGND / Sensor GND	Sensor ground.
ECU	Electronic Control Unit: engine management computer.
SRAM	ECU working memory, where the live data is held.
CAN	Communication bus between the gauge and the ECU (CAN H and CAN L lines).
SSR	Solid State Relay.
AFR	Air/Fuel Ratio (14.7 = stoichiometric on petrol).
Wideband	Wideband lambda (oxygen) sensor.

Term	Meaning
MAP	Manifold Absolute Pressure sensor.
IAT	Intake Air Temperature.
TPS	Throttle Position Sensor.
APC	Saab Automatic Performance Control: boost pressure control valve.
NTC	Resistor whose value drops as temperature rises (temperature sensor).
EGT	Exhaust Gas Temperature.
Knock	Detonation detected by the ECU.
Misfire	Ignition misfire.
Log	Data recording on the SD card.